Determinants

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle 4, 3\rangle, \langle 2, 5\rangle$.

$4(5) - 3(2) = 20 - 6 = 14$.

2 **[Foundational] • Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 1, 2, 3\rangle, \langle 0, 1, 4\rangle, \langle 5, 6, 0\rangle$ by cofactor expansion along the first row.

$1\det(\langle 1, 4\rangle, \langle 6, 0\rangle) - 2\det(\langle 0, 4\rangle, \langle 5, 0\rangle) + 3\det(\langle 0, 1\rangle, \langle 5, 6\rangle) = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1$.

3 **[Foundational] • Target: 3x3 determinants via cofactor expansion**

Recompute the determinant of the matrix with rows $\langle 1, 2, 3\rangle, \langle 0, 1, 4\rangle, \langle 5, 6, 0\rangle$ by expanding along the SECOND row instead, to confirm the answer matches.

Expanding along row 2 (with alternating signs $-, +, -$): $-0\det(\langle 2, 3\rangle, \langle 6, 0\rangle) + 1\det(\langle 1, 3\rangle, \langle 5, 0\rangle) - 4\det(\langle 1, 2\rangle, \langle 5, 6\rangle) = 0 + 1(0 - 15) - 4(6 - 10) = -15 + 16 = 1$. Matches the earlier result of 1.

4 **[Foundational] • Target: The determinant product rule**

For A with rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ and B with rows $\langle 2, 1\rangle, \langle 0, 3\rangle$, verify $\det(AB) = \det(A)\det(B)$.

$\det A = 4 - 6 = -2$, $\det B = 6 - 0 = 6$, so $\det A \det B = -12$. AB has rows $\langle 2, 7\rangle, \langle 6, 15\rangle$ (computed as usual matrix multiplication), and $\det(AB) = 2(15) - 7(6) = 30 - 42 = -12$. They match.

5 **[Foundational] • Target: The determinant product rule**

Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \frac{1}{\det A}$ for an invertible matrix A.

Since $AA^{-1} = I$ and $\det I = 1$, the product rule gives $\det(A)\det(A^{-1}) = 1$. Dividing both sides by $\det A$ (nonzero since A is invertible) gives $\det(A^{-1}) = \frac{1}{\det A}$.

6 **[Foundational] • Target: Determinants and scalar multiples / row operations**

If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).

For an $n \times n$ matrix, $\det(cA) = c^n \det A$. For $n = 2$: $\det(2A) = 2^2(5) = 20$.

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7 [Foundational] • Target: Determinants and scalar multiples / row operations

If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.

$\det(2A) = 2^3(5) = 40$ (note this is different from the 2×2 case — the exponent depends on the matrix's size).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinants and scalar multiples / row operations

Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2\rangle$, $\langle 3, 4\rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4\rangle$, $\langle 1, 2\rangle$.

Swapping rows negates the determinant: $\det = -(-2) = 2$. (Direct check: $3(2) - 4(1) = 6 - 4 = 2$.)

9 [Intermediate] • Target: The invertibility criterion

State the invertibility criterion in terms of the determinant.

A square matrix A is invertible if and only if $\det A \neq 0$.

10 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 2, 4\rangle$, $\langle 1, 2\rangle$ invertible? Justify using the determinant.

$\det = 2(2) - 4(1) = 0$. Since the determinant is 0, the matrix is NOT invertible (singular).

11 [Intermediate] • Target: Applying the invertibility criterion

For what value(s) of k is the matrix with rows $\langle k, 2\rangle$, $\langle 3, k\rangle$ NOT invertible?

$\det = k^2 - 6$. Setting $k^2 - 6 = 0$ gives $k = \pm\sqrt{6}$. The matrix fails to be invertible exactly when $k = \sqrt{6}$ or $k = -\sqrt{6}$.

12 [Intermediate] • Target: Geometric meaning of the determinant

For a 2×2 matrix A with rows $\mathbf{r}_1$, $\mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0\rangle$, $\mathbf{r}_2 = \langle 1, 4\rangle$.

$\det = 3(4) - 0(1) = 12$. Area $= |12| = 12$.

13 [Intermediate] • Target: Geometric meaning of the determinant

Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.

Two identical rows span a degenerate "parallelogram" with zero area (the two vectors are parallel, in fact equal, so they don't sweep out any two-dimensional region) — matching $\det = 0$.

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14 [Intermediate] • Target: Determinants of special matrices

Find the determinant of the identity matrix I_3 (rows $\langle 1, 0, 0\rangle$, $\langle 0, 1, 0\rangle$, $\langle 0, 0, 1\rangle$).

$\det I_3 = 1$ (expanding along any row/column, the product of the diagonal entries, all 1's, with every off-diagonal cofactor contributing 0).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the determinant of the matrix with rows $\langle 3, 0, 0\rangle$, $\langle 0, -2, 0\rangle$, $\langle 0, 0, 5\rangle$ (a diagonal matrix).

For a diagonal (or more generally triangular) matrix, the determinant is the product of the diagonal entries: $3(-2)(5) = -30$.

16 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the determinant of the matrix with rows $\langle 0, 0, 3\rangle$, $\langle 0, 1, 0\rangle$, $\langle 3, 0, 0\rangle$, and explain why it is nonzero even though the main diagonal has two zeros.

Expanding along row 2 (only nonzero entry is the middle 1): $\det = 1 \cdot \det(\langle 0, 3\rangle, \langle 3, 0\rangle) = 1(0 - 9) = -9$. The main-diagonal shortcut only applies to TRIANGULAR matrices; this matrix is not triangular (it has nonzero entries above AND below the diagonal), so the diagonal-product rule doesn't apply here.

17 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the determinant of the zero matrix (all entries 0) of any size, and connect this to the invertibility criterion.

$\det = 0$ for the zero matrix of any size (every row is the zero vector, so every cofactor expansion term vanishes). This matches the invertibility criterion: the zero matrix is never invertible.

18 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Explain why any matrix with an entire row (or column) of zeros must have determinant 0, using cofactor expansion along that row.

Expanding along a row of all zeros, every term in the cofactor expansion is (zero entry)$\times$(its cofactor) $= 0$, so the whole sum is 0.

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19 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

Cramer's Rule solves $ax + by = e$, $cx + dy = f$ via $x = \frac{\det A_x}{\det A}$, $y = \frac{\det A_y}{\det A}$, where A is the coefficient matrix and A_x, A_y replace the respective column with $\langle e, f \rangle$. Use it to solve $2x + 3y = 8$, $x - y = 1$.

A has rows $\langle 2, 3 \rangle$, $\langle 1, -1 \rangle$, $\det A = -2 - 3 = -5$. A_x has rows $\langle 8, 3 \rangle$, $\langle 1, -1 \rangle$, $\det A_x = -8 - 3 = -11$, so $x = \frac{-11}{-5} = \frac{11}{5}$. A_y has rows $\langle 2, 8 \rangle$, $\langle 1, 1 \rangle$, $\det A_y = 2 - 8 = -6$, so $y = \frac{-6}{-5} = \frac{6}{5}$.

20 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 2.01 Systems of Linear Equations

Explain why Cramer's Rule fails (gives no unique solution) whenever $\det A = 0$.

Division by $\det A = 0$ is undefined, and this matches the invertibility criterion: $\det A = 0$ means the coefficient matrix is singular, so the system does not have a unique solution (it has either no solution or infinitely many).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: 2x2 determinants

Find the slope of the least-squares line for $(-1, -1)$, $(3, 2)$, $(4, 1)$.

a. $-\frac{1}{3}$

b. $\frac{3}{2}$

c. $\frac{1}{2}$ ✓

d. $-\frac{1}{2}$

Normal equations: $[(3, 6; 6, 26)][\beta_0; \beta_1] = [2; 11]$, giving $\hat{\beta}_0 = -\frac{1}{3}$, $\hat{\beta}_1 = \frac{1}{2}$.

22 [Foundational] • Target: 3x3 determinants via cofactor expansion

Given $A = [(-8, 9; -6, 7)] = PDP^{-1}$ with $P = [(3, 1; 2, 1)]$, $D = [(-2, 0; 0, 1)]$, find the (2,2) entry of A^4.

a. -29 ✓

b. 2401

c. 19

d. -28

$A^4 = PD^4P^{-1}$ with $D^4 = [(16, 0; 0, 1)]$. Multiplying out, $A^4 = [(46, -45; 30, -29)]$, so the entry is -29.

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23 [Foundational] • Target: The determinant product rule

Find the distance from the point $(4,\ -1)$ to the line through the origin spanned by $\langle 2,\ -2\rangle$.

a. $\frac{5\sqrt{2}}{2}$

b. $\sqrt{17}$

c. $\frac{3\sqrt{2}}{2}$ ✓

d. $3\sqrt{2}$

The distance is $||\mathbf{y} - \text{proj}_{\mathbf{u}}\mathbf{y}|| = ||\langle \frac{3}{2}, \frac{3}{2}\rangle|| = \frac{3\sqrt{2}}{2}$.

24 [Foundational] • Target: Determinants and scalar multiples / row operations

Find the orthogonal projection of $\mathbf{y} = \langle -4,\ -2\rangle$ onto $\mathbf{u} = \langle 0, 2\rangle$.

a. $\langle -4, 0\rangle$

b. $\langle \frac{4}{5}, \frac{2}{5}\rangle$

c. $\langle 0,\ -2\rangle$ ✓

d. $\langle 0,\ -8\rangle$

$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u} = \frac{-4}{4}\mathbf{u} = \langle 0,\ -2\rangle$.

25 [Foundational] • Target: The invertibility criterion

Which vector is an eigenvector of $A = [(3,\ -12; 2,\ -7)]$ for $\lambda = -3$?

a. $\langle -3,\ -1\rangle$

b. $\langle 1, 2\rangle$

c. $\langle 2, 1\rangle$ ✓

d. $\langle 2,\ -1\rangle$

Solve $(A - \ -3I)\mathbf{v} = \mathbf{0}$: $[(6,\ -12; 2,\ -4)]$ has null space spanned by $\langle 2, 1\rangle$.

26 [Foundational] • Target: Applying the invertibility criterion

What is the dimension of $\text{span}\{x^2 + 1, 2x^2 + 2\}$ in P_2?

a. 2

b. 1 ✓

c. 3

d. 0

Write each polynomial as a coefficient vector in $\{1, x, x^2\}$ and row-reduce: 1 of them are independent, so the dimension is 1.

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27 [Foundational] • Target: Geometric meaning of the determinant

Find the orthogonal projection of $\langle -1, 4, 3\rangle$ onto $W = \text{span}\{\langle 1, 0, 1\rangle, \langle 1, 0, -1\rangle\}$ (an orthogonal basis).

a. $\langle 0, 4, 0\rangle$

b. $\langle 1, 0, 1\rangle$

c. $\langle -1, 0, 3\rangle$ ✓

d. $\langle -2, 0, 6\rangle$

$\hat{\mathbf{y}} = \sum \frac{\mathbf{y}\cdot\mathbf{u}_i}{\mathbf{u}_i\cdot\mathbf{u}_i}\mathbf{u}_i = \langle -1, 0, 3\rangle$.

28 [Foundational] • Target: Determinants of special matrices

Find the characteristic polynomial of $[(3, -4; -4, -2)]$.

a. $\lambda^2 - \lambda - 22$ ✓

b. $\lambda^2 + \lambda - 22$

c. $\lambda^2 - \lambda + 22$

d. $\lambda^2 + 22\lambda + 1$

$\det(A - \lambda I) = \lambda^2 - (\text{tr}A)\lambda + \det A$ with $\text{tr}A = 1$, $\det A = -22$: $\lambda^2 - \lambda - 22$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cramer's Rule

$\langle 1, 1, 1\rangle, \langle 1, -1, 0\rangle, \langle 1, 1, -2\rangle$ is an orthogonal basis. Find the coefficient of $\langle 1, 1, -2\rangle$ when $\langle 5, 2, -5\rangle$ is written in this basis.

a. 17

b. $\frac{17}{3}$

c. $-\frac{17}{6}$

d. $\frac{17}{6}$ ✓

For an orthogonal basis $c_i = \frac{\mathbf{y}\cdot\mathbf{u}_i}{\mathbf{u}_i\cdot\mathbf{u}_i} = \frac{17}{6} = \frac{17}{6}$.

30 [Intermediate] • Target: 2x2 determinants

A has eigenvalues $2, 3$. What are the eigenvalues of $A + 2I$?

a. $2, 3$

b. $4, 6$

c. $4, 5$ ✓

d. $-4, -5$

If $A\mathbf{v} = \lambda\mathbf{v}$ then $A + 2I$ acts on $\mathbf{v}$ by the same function of λ, giving $4, 5$.

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31 [Intermediate] • Target: 3x3 determinants via cofactor expansion

Are $\langle 1, 4\rangle$ and $\langle -2, -8\rangle$ linearly independent?

a. Yes, they are independent

b. Only if one of them is the zero vector

c. It cannot be decided from the vectors alone

d. **No, they are dependent** ✓

$\det[(1, -2; 4, -8)] = 0$. A zero determinant means one vector is a multiple of the other.

32 [Intermediate] • Target: The determinant product rule

Find the image of $\langle 3, 1\rangle$ under the reflection across the x-axis.

a. $\langle -3, 1\rangle$

b. $\langle 3, 1\rangle$

c. $\langle -1, 3\rangle$

d. $\langle 3, -1\rangle$ ✓

The standard matrix is $[(1, 0; 0, -1)]$; multiplying gives $\langle 3, -1\rangle$.

33 [Intermediate] • Target: Determinants and scalar multiples / row operations

Find the image of $\langle 1, -2\rangle$ under the scaling by the factor 3.

a. $\langle -3, 6\rangle$

b. $\langle 1, -2\rangle$

c. $\langle -6, 3\rangle$

d. $\langle 3, -6\rangle$ ✓

The standard matrix is $[(3, 0; 0, 3)]$; multiplying gives $\langle 3, -6\rangle$.

34 [Intermediate] • Target: The invertibility criterion

Find the image of $\langle -1, 2\rangle$ under the orthogonal projection onto the x-axis.

a. $\langle -1, 0\rangle$ ✓

b. $\langle 1, 0\rangle$

c. $\langle -1, 2\rangle$

d. $\langle 0, -1\rangle$

The standard matrix is $[(1, 0; 0, 0)]$; multiplying gives $\langle -1, 0\rangle$.

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35 [Intermediate] • Target: Applying the invertibility criterion

Find the image of $\langle 3, 2\rangle$ under the horizontal shear x ■ x + 3y.

a. $\langle 3, 11\rangle$

b. $\langle 9, 2\rangle$ ✓

c. $\langle -9, -2\rangle$

d. $\langle 3, 2\rangle$

The standard matrix is $[(1, 3; 0, 1)]$; multiplying gives $\langle 9, 2\rangle$.

36 [Intermediate] • Target: Geometric meaning of the determinant

What is the dimension of span of $\langle 2, -3, 1\rangle$, $\langle 2, -3, 1\rangle$, $\langle 2, -3, 1\rangle$, $\langle -4, 6, -2\rangle$?

a. 4

b. 2

c. 1 ✓

d. 3

Row-reduce the matrix with these vectors as rows; it has 1 nonzero row(s), so the span has dimension 1.

37 [Intermediate] • Target: Determinants of special matrices

Let W be spanned by $\langle 8, -2, 4, 0, 3\rangle$, $\langle -2, 0, -2, 0, -3\rangle$, $\langle -1, 0, -2, -2, 3\rangle$, $\langle 3, -1, 1, 0, 0\rangle$ in R^5. What is $\dim W^{\perp}$?

a. 3

b. 5

c. 2 ✓

d. 4

$W^{\perp}$ is the null space of the matrix with these rows. Its rank is 3, so $\dim W^{\perp} = 5 - 3 = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the dimension of the null space of $[(3, 3, -2, 1, -1, -3; 0, 0, 0, 0, 0, 0; -3, -3, 2, -1, 1, 3; -3, -3, 2, -1, 1, 3)]$.

a. 1

b. 5 ✓

c. 3

d. 6

Row reduction shows rank 1. By rank–nullity, $\dim \mathrm{Nul} A = n - \mathrm{rank} A = 6 - 1 = 5$.

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39 [Challenging] • Target: 2x2 determinants • Cross-Lesson: 1.01 Vectors in n Dimensions

What is the dimension of the subspace of R^4 defined by $2x_1 - x_2 + 2x_3 - 2x_4 = 0,\ -x_1 - 3x_2 - x_4 = 0$?

a. 2 ✓

b. 4

c. 3

d. 1

The 2 equation(s) are independent (rank 2), so the solution space has dimension $4 - 2 = 2$.

40 [Challenging] • Target: 3x3 determinants via cofactor expansion • Cross-Lesson: 2.01 Systems of Linear Equations

Find the eigenvalues of $[(4, 0; 3, 3)]$.

a. $0, 4$

b. $4, 5$

c. $-3, -4$

d. $3, 4$ ✓

The matrix is triangular, so its eigenvalues are its diagonal entries: $3, 4$.

41 [Challenging] • Target: The determinant product rule • Cross-Lesson: 3.01 Matrices & Matrix Algebra

$\mathbf{v} = \langle 4, 6\rangle$ is an eigenvector of $A = [(7, -2; 6, 0)]$. What is its eigenvalue?

a. 3

b. 5

c. 4 ✓

d. -4

$A\mathbf{v} = \langle 16, 24\rangle = 4\langle 4, 6\rangle$, so $\lambda = 4$.

42 [Challenging] • Target: Determinants and scalar multiples / row operations • Cross-Lesson: 1.01 Vectors in n Dimensions

$T : R^7 R^5$ is linear with range of dimension 4. What is $\dim \ker T$?

a. 3 ✓

b. 1

c. 4

d. 7

Rank–nullity: $\dim \ker T + \dim \text{range} T = 7$, so $\dim \ker T = 3$.

43 [Challenging] • Target: The invertibility criterion • Cross-Lesson: 2.01 Systems of Linear Equations

What is the dimension of $\{A \in M_{2\times 2} : \text{tr} A = 0\}$?

a. 3 ✓

b. 4

c. 2

d. 6

The dimension is 3: one linear condition removes one dimension.

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44 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For which k are $\langle 0, -2, -2\rangle$ and $\langle -1, 2, k\rangle$ orthogonal?

a. 2

b. -2 ✓

c. -1

d. 3

Set the dot product to zero: $-2k - 4 = 0$, so $k = -2$.

45 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 1.01 Vectors in n Dimensions

$T : R^6 R^4$ is linear with $\dim \ker T = 3$. What is the dimension of its range?

a. 3 ✓

b. 4

c. 1

d. 6

Rank–nullity: $\dim \text{range} T = 6 - 3 = 3$.

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